

EXPERIENCE

UST – Data Engineer | 12/2022 - Present

Blue Cross Blue Shield Association - National Consumer Cost Tool

- Facilitated the AWS Cloud Migration of an Informatica and SOLR application that allowed members to view cost information of healthcare providers in their area, increasing scalability and performance of data workflows while optimizing cost.
- Built data pipelines by developing PySpark scripts that were used to extract, structure, and validate millions of record-level data from zip files sent to S3, along with generating reports for performance metrics, data validation, and data quality. Created PySpark scripts for data processing and transformation of supporting datasets.
- Designed the ETL process by integrating AWS services including S3, Lambda, Glue and Athena along with PySpark scripts to ingest cleaned datasets into OpenSearch to support API Development. Created Iceberg Tables in AWS Data Catalog to allow for SQL querying through Athena for Analytics Team.
- Provisioned Cloud Infrastructure by creating and maintaining Infrastructure-as-Code through CloudFormation. Contributed to implementation of AWS CodePipeline, automating deployment of cloud environment.
- Supported cross-functional teams by optimizing performance in higher environments, resolving file failures, addressing data inconsistencies, and implementing enhancements, improving the reliability and stability of the ETL pipeline.
- Optimized data workflows by developing reusable components, implementing error handling, and automating alerts, ensuring easier resolution of issues while allowing for future scalability.
- Utilized Agile methodologies with Jira software by reviewing business requirements, tracking progress of cloud migration, and presenting key milestones to stakeholders.

Blue Cross Blue Shield Association – Provider Engagement Data Exchange Platform

- Automated testing for the AWS Cloud Migration of a Hadoop application which utilized Hive, Apache NiFi, and HDFS to transfer PHI/PII healthcare data between different entities for reimbursement and analytical purposes.
- Wrote Python scripts to streamline the testing process by creating synthetic member information, generating thousands of data requests, and performing comparisons between JSONs.
- Built Lambda Functions in Python with AWS EventBridge to automate daily data backup and stopping EC2 instances at regular intervals.
- Ensured the cloud application maintained data integrity and accuracy while improving application performance. Identified performance bottlenecks, errors, and discrepancies between the cloud and legacy application. Executed SQL queries with Athena and Hive queries on legacy platform to verify proper data transfer and file submissions.

CERTIFICATIONS

- dbt Labs – dbt Fundamentals, 01/2025
- Amazon Web Services - Certified Cloud Practitioner, 01/2024

SKILLS

- **Programming:** Python, SQL, Shell/Bash Scripting
- **Big Data Technologies/Data Engineering:** Apache Spark, NiFi, and Hadoop
- **Cloud Technologies (AWS):** EC2, IAM, Lambda, Glue, CloudFormation and CodePipeline
- **Data Storage:** HDFS, Hive, Elasticsearch, Iceberg, PostgreSQL, SQL Server, S3, OpenSearch and Athena
- **Data Analysis and Visualization:** Tableau, Jupyter Notebook (Pandas, NumPy, Seaborn, and Matplotlib)
- **Command Line:** Linux/Unix
- **Version Control and Project Management:** Git, GitHub, and Jira

EDUCATION

North Park University, Chicago, IL

Bachelor's in Biomedical Science, Graduation Year 2021, 3.69/4.0 GPA

Dean's List, Platinum Scholarship, Phi Theta Kappa Honors